Exp 1-Design, Realization and Performance Analysis of Butterworth IIR Low-Pass and High-Pass Digital Filters Using MATLAB

Program:

```
clc;
clear;
close all;

%% ============================================================
% BUTTERWORTH IIR DIGITAL FILTER DESIGN
% LOW-PASS AND HIGH-PASS
% ============================================================

%% Input Specifications
Fs = 1000;          % Sampling frequency (Hz)

Ap = 1;             % Passband attenuation (dB)
Ast = 40;            % Stopband attenuation (dB)

% Low-Pass specifications
Fp_LP  = 100;       % Passband frequency (Hz)
Fst_LP = 150;       % Stopband frequency (Hz)

% High-Pass specifications
Fst_HP = 200;       % Stopband frequency (Hz)
Fp_HP  = 250;       % Passband frequency (Hz)


%% ============================================================
% BUTTERWORTH LOW-PASS FILTER
% ============================================================

Wp_LP = Fp_LP/(Fs/2);
Ws_LP = Fst_LP/(Fs/2);

[N_LP, Wn_LP] = buttord(Wp_LP, Ws_LP, Ap, Ast);

[b_LP, a_LP] = butter(N_LP, Wn_LP, 'low');


%% ============================================================
% BUTTERWORTH HIGH-PASS FILTER
% ============================================================

Ws_HP = Fst_HP/(Fs/2);
Wp_HP = Fp_HP/(Fs/2);

[N_HP, Wn_HP] = buttord(Wp_HP, Ws_HP, Ap, Ast);

[b_HP, a_HP] = butter(N_HP, Wn_HP, 'high');


%% ============================================================
% DISPLAY FILTER PARAMETERS
% ============================================================

fprintf('\n=======================================\n');
```

```
fprintf(' BUTTERWORTH IIR FILTER DESIGN\n');
fprintf('=======================================\n');

fprintf('\nLOW-PASS FILTER\n');
fprintf('Filter Order = %d\n', N_LP);
fprintf('Normalized Cutoff = %.4f\n', Wn_LP);

fprintf('\nNumerator Coefficients (b):\n');
disp(b_LP);

fprintf('Denominator Coefficients (a):\n');
disp(a_LP);


fprintf('\nHIGH-PASS FILTER\n');
fprintf('Filter Order = %d\n', N_HP);
fprintf('Normalized Cutoff = %.4f\n', Wn_HP);

fprintf('\nNumerator Coefficients (b):\n');
disp(b_HP);

fprintf('Denominator Coefficients (a):\n');
disp(a_HP);


%% ============================================================
% FREQUENCY RESPONSE
% ============================================================

[H_LP, f_LP] = freqz(b_LP, a_LP, 2048, Fs);
[H_HP, f_HP] = freqz(b_HP, a_HP, 2048, Fs);


%% ============================================================
% SINGLE OUTPUT WINDOW
% ============================================================

figure('Name','Butterworth IIR Filter Analysis',...
       'NumberTitle','off');

% ---- LPF Magnitude ----
subplot(3,2,1);
plot(f_LP,20*log10(abs(H_LP)),'LineWidth',1.2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Butterworth LPF - Magnitude');
xline(Fp_LP,'--r');
xline(Fst_LP,'--k');

% ---- LPF Phase ----
subplot(3,2,2);
plot(f_LP,unwrap(angle(H_LP)),'LineWidth',1.2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('Butterworth LPF - Phase');

% ---- HPF Magnitude ----
subplot(3,2,3);
```

```
plot(f_HP,20*log10(abs(H_HP)),'LineWidth',1.2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Butterworth HPF - Magnitude');
xline(Fst_HP,'--r');
xline(Fp_HP,'--k');

% ---- HPF Phase ----
subplot(3,2,4);
plot(f_HP,unwrap(angle(H_HP)),'LineWidth',1.2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('Butterworth HPF - Phase');

% ---- LPF Pole-Zero ----
subplot(3,2,5);
zplane(b_LP,a_LP);
title('Butterworth LPF - Pole-Zero');

% ---- HPF Pole-Zero ----
subplot(3,2,6);
zplane(b_HP,a_HP);
title('Butterworth HPF - Pole-Zero');


%% ============================================================
% STABILITY ANALYSIS
% ============================================================

poles_LP = roots(a_LP);
poles_HP = roots(a_HP);

fprintf('\n========================================\n');
fprintf(' STABILITY ANALYSIS\n');
fprintf('========================================\n');

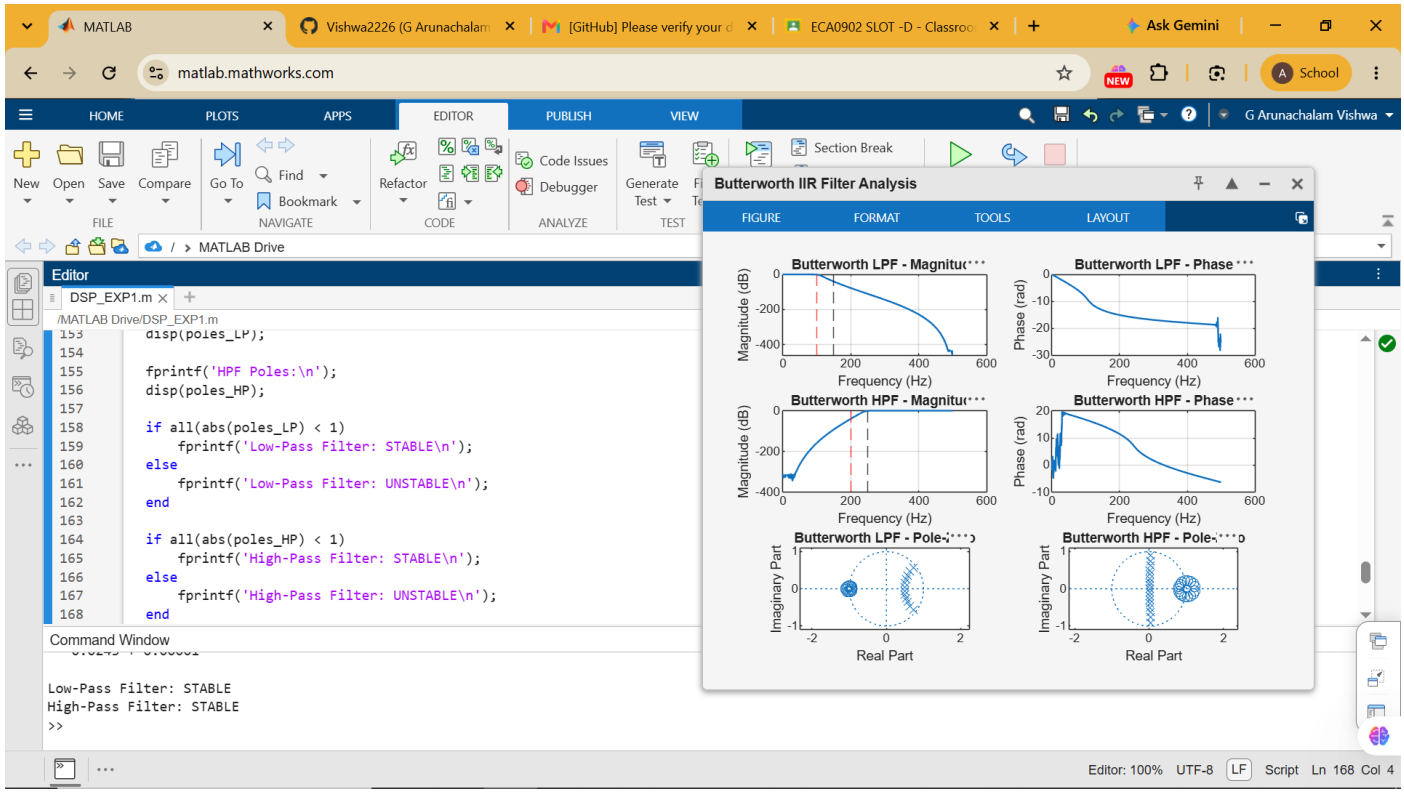
fprintf('\nLPF Poles:\n');
disp(poles_LP);

fprintf('HPF Poles:\n');
disp(poles_HP);

if all(abs(poles_LP) < 1)
    fprintf('Low-Pass Filter: STABLE\n');
else
    fprintf('Low-Pass Filter: UNSTABLE\n');
end

if all(abs(poles_HP) < 1)
    fprintf('High-Pass Filter: STABLE\n');
else
    fprintf('High-Pass Filter: UNSTABLE\n');
end
```

Output:


MATLAB
matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
Butterworth IIR Filter Analysis
FIGURE
FORMAT
TOOLS
LAYOUT
Butterworth LPF - Magnitude
Butterworth LPF - Phase
Butterworth HPF - Magnitude
Butterworth HPF - Phase
Butterworth LPF - Pole-Zero
Butterworth HPF - Pole-Zero
Magnitude (dB)
Phase (rad)
Frequency (Hz)
Imaginary Part
Real Part
Editor
DSP_EXP1.m
/MATLAB Drive/DSP_EXP1.m
fprintf('HPF Poles:\n');
disp(poles_HP);
if all(abs(poles_LP) < 1)
fprintf('Low-Pass Filter: STABLE\n');
else
fprintf('Low-Pass Filter: UNSTABLE\n');
end
if all(abs(poles_HP) < 1)
fprintf('High-Pass Filter: STABLE\n');
else
fprintf('High-Pass Filter: UNSTABLE\n');
end
Command Window
Low-Pass Filter: STABLE
High-Pass Filter: STABLE
>>
Editor: 100%
UTF-8
Script
Ln 168 Col 4